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Adjoint-Based Functional Correction for Unstructured Mesh Finite Volume Methods

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Abstract The goal of this paper is to demonstrate the use of adjoint-based functional correction and mesh adaptation for aerodynamic flows with an unstructured mesh finite volume solver. A key feature of our approach is that all calculations are performed on a single mesh, unlike other error correction and mesh adaptation schemes. As using the original p-truncation error estimate is not helpful in improving the functional, we use a smoothed estimate of the truncation error to correct the functional for both inviscid and viscous flows. The correction term is based on the smoothed truncation error and the adjoint solution, with both the continuous and discrete adjoints. The same correction term is used as an adaptation indicator for goal-based mesh adaptation as well. Our results show the effectiveness of our method in improving the convergence rate for test cases of interest in computational aerodynamics.

Keywords Adjoint-based functional correction · Adjoint-based mesh adaptation · Finite volume method · Unstructured mesh · Truncation error estimation

List of symbols

A	Jacobian matrix
f	Source term of the primal problem
g	Source term of the adjoint problem
J	Output functional

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α	Angle of attack
Ma	Mach number
Re	Reynolds number
τ	Visous stress term

1 Introduction

Accurate computational simulations of complex aerodynamic flows have become a key component of the design process in the aerospace industry. As the reliance on computer simulation has increased, greater emphasis is being placed on quantifying the accuracy of the simulation results. Moreover, the complexity and the scale of problems simulated using computational aerodynamics has grown implying that besides accuracy and reliability, the efficiency of the simulation is also crucial. The desire for reliable, efficient computational algorithms has motivated a great deal of research during the last years. For efficiency, the output functional value should be computed to a higher order of accuracy than the overall solution on which it is based. For reliability, a bound on the remaining error in the functional should be computed. Adjoint-based techniques can be used to estimate the error in an output functional, such as lift or drag, and provide local indicators for adaptive methods. Within the context of finite element methods, output functional correction has been outlined by Becker and Rannacher [1] for a posteriori error estimation for elliptic, parabolic, and hyperbolic equations and by Larson and Barth [2] based on structural finite element methods [3]. Machiels and co-workers [4] have also established an implicit a posteriori procedure for output error estimation for finite element method. Pierce and Giles have described the adjoint correction technique [5] and extended the inherent super-convergence properties of finite element methods to cover numerical results obtained from finite difference and finite volume without natural super-convergence properties. Moreover, the technique can be used to improve the accuracy of super-convergent functionals obtained from finite element methods by constructing smoother, higher-order interpolants of the primal and adjoint solutions [6]. This method for output error estimation and correction for two-dimensional flows was applied by Venditti and Darmofal for a second-order finite volume discretization for both inviscid [7] and viscous flows [8]. They also presented an error estimation strategy based on the adjoint formulation for estimating errors in functional outputs for one-dimensional problems and their error estimation procedure was applied to a standard, second-order finite volume discretization [9].

We will show later, see Eq. 16, that for a given primal problem, $LU = f$, and the adjoint problem, $L^*Z = g$, the output functional based on the discrete solution can be written as

$$J_h = \left(L^* Z_h, U_h\right)_D + \left(B^* Z_h, C U_h\right)_{\partial D} + \left(Z_h, L\left(U - U_h\right)\right)_D \tag{1}$$

where the first and second term are the volume integral and the boundary integral of the output functional, respectively and the last term is the error in calculating the output functional. The error is expressed as an inner product of the local residual errors and adjoint variables. So the error in a chosen functional can be directly related to the local truncation error of the primal solution through the adjoint variable. Truncation error is defined as the amount by which a discrete solution fails to satisfy the PDE locally. While truncation error analysis for structured mesh schemes is well defined as a result of Taylor series analysis, the analysis of unstructured mesh schemes has lagged behind. Ollivier-Gooch and Van-Altena [10] performed Taylor series truncation error analysis for the Laplacian on regular triangular meshes, and this was extended to general stencils by Jalali and Ollivier-Gooch [11]. The output of this analysis is

a set of Taylor series coefficients for the truncation error for each control volume which can be combined with local solution derivatives to compute the truncation error. Although the truncation error and discretization error[1] have the same asymptotic order of accuracy and both have smooth distributions for structured meshes, the truncation error for unstructured finite volume schemes is asymptotically larger than the discretization error [11–13]. Furthermore, both have a noisy appearance as a consequence of discontinuous jump of the coefficients of the terms in the Taylor series expansion of the error from one control volume to another. Sharbatdar and Ollivier-Gooch [14] have shown by eigenanalysis of the truncation error that the rough modes dominate the unstructured mesh truncation error.

Output error correction based on the original estimate of the truncation error with rough modes is not helpful as we will show in our results, and therefore, we developed a continuous truncation error estimate by producing a continuous approximation to the finite volume discrete solution and estimating truncation error as the higher-order residual of the PDE for this continuous function on the same mesh. Therefore, the truncation error is estimated using a single mesh scheme. This continuous interpolation of the discrete solution is used to compute the truncation error which is multiplied by the adjoint solution in Eq. 1 as the correction term. The effectiveness of this smoothed truncation error in correcting the output functional has been shown for a variety of problems with manufactured solutions where the exact functional can be calculated [15]. The main goal of this paper is to demonstrate how successful the smoothed estimate of the truncation error is in correcting the functionals for general problems which are of interest in computational aerodynamics. These flow test cases do not have an exact solution and we are considering flows around more complex geometries.

Furthermore, the smoothed estimate of the truncation error is used for mesh adaptation purposes in this paper and the effectiveness of the smoothed truncation error we developed is examined for mesh adaptation.

Mesh adaptation seeks to automate the process of minimizing discretization error by first computing the solution on a coarse mesh and then successively refining the mesh so that the optimal mesh for the solution being computed is produced [1,7–9,16–19]. We will show how successful the smoothed truncation error is in adapting the mesh to improve the functional. The smoothed truncation error weighted by the adjoint solution is used as the adaptation indicator.

We begin by shortly describing our higher order finite volume flow solver. The two applications described in this paper are functional correction, Sect. 4, and mesh adaptation, Sect. 5. Both these applications are dependent on the adjoint solution, explained in Sect. 3 and the truncation error. The smoothed estimate of the truncation error we are using is illustrated in Sect. 6. The error estimation process is done by solving the primal and adjoint problems at second-order. A variety of functional correction and mesh adaptation results are shown for inviscid and viscous flows in Sect. 7.

2 Higher Order Finite Volume Flow Solver

The finite volume formulation of a general conservative PDE can be written as:

$$\iint_{CV} \frac{\partial \vec{U}}{\partial t} dA + \oint_{CS} \vec{F} \cdot \hat{n}\, ds = 0 \tag{2}$$

[1] Discretization error is the difference between the exact solution and the discrete solution.

where $\overrightarrow{U}$ denotes the solution vector, $\overrightarrow{F}$ is the flux vector, $\hat{n}$ represents the outward unit normal vector, CV and CS are the control volume and control surface (the boundary of the control volume), respectively, and ds is the infinitesimal length along the control surface. For a fixed mesh, the time derivative can be brought out from the integral in Eq. 2:

$$\frac{d\overline{U}_i}{dt} = -\frac{1}{A_{CV_i}} \oint_{CS_i} \vec{F} \cdot \hat{n}\, ds \equiv -R_i\left(\overline{U}_i\right) \tag{3}$$

The right hand side is called the flux integral or residual and represents the spatial discretization of the same control volume. To compute the flux integral for each control volume, the numerical flux is computed at Gauss quadrature points based on the reconstructed solution. For second-order, one quadrature point is required and for third and fourth order flux integration, two quadrature points per edge are necessary. To obtain a single flux from two different values at each Gauss point, we use Roe's scheme [20] for inviscid fluxes; for viscous fluxes, we average the gradients and add a jump term [21,22].

For steady-state problems, $R\left(\bar{U}_i\right) = 0$. This non-linear system of equations could be solved by the direct application of Newton's methods. However, Newton's method will diverge if the initial guess is too far from the real solution. Instead, we use implicit time-stepping with a local time step Δt_i [23], in which the next time level solution for both the space and the time discretizations are used. Denoting the solution average vector at the current time level n by $\overline{U}_i^n$, both sides of Eq. 3 should be evaluated at the next time level $n+1$. After linearization, we get

$$\left(\frac{I}{\Delta t_i} + \frac{\partial R}{\partial \overline{U}}\right)\delta U_i = -R_i\left(\overline{U}_i^n\right), \qquad \overline{U}_i^{n+1} = \overline{U}_i^n + \delta U_i \tag{4}$$

where $\frac{\partial R}{\partial \overline{U}}$ is the Jacobian matrix resulting from flux integral linearization. Equation 4 is a large linear system of equations which needs to be solved at each time step to obtain an update for the vector of unknowns. We evaluate the global Jacobian matrix explicitly using the chain rule [23–25].

3 Discrete and Continuous Adjoint Methods

The adjoint theory in the context of linear algebra is the basis for the discrete adjoint approach. In this approach, the algebraic equations arising from discretization of the original fluid dynamic equations are used. Alternatively, the adjoint PDE can be formulated and discretized independently, which is the basis of the continuous adjoint formulation.

3.1 Discrete Adjoint

Consider a partial differential equation arising from a finite volume discretization of the original fluid dynamic equations which is discretized and written as an algebraic equation:

$$R_h\left(\overline{U}_h\right) = 0 \tag{5}$$

For a scalar output, $J_h\left(\overline{U}_h\right)$ such as lift or drag, the associated discrete adjoint vector, Z_h, must satisfy the discrete adjoint equation:

$$\left(\frac{\partial R_h}{\partial \overline{U}_h}\right)^T Z_h = -\left(\frac{\partial J_h}{\partial \overline{U}_h}\right)^T \tag{6}$$

A detailed description of the discrete adjoint formulation is explained by Fidkowski and Darmofal [26]. Because our finite volume solver can compute the Jacobian matrix, we can formulate the discrete adjoint problem directly as a linear system. The discrete adjoint formulation of Eq. 6 is independent of the governing equations; also, the only part that differs between cases for a given governing equation is the sensitivity of the output functional with respect to the primal problem.

3.2 Continuous Adjoint

The continuous primal problem can be posed as

$$\begin{array}{rlll} \text{Determine} & J & = (U, g)_D + (CU, h)_{\partial D} & \\ \text{given that} & LU & = f & in\ D \\ & BU & = e & in\ \partial D \end{array} \tag{7}$$

where $(U, V)_D = \int_D U^T V dv$ is an integral over the domain and $(U, V)_{\partial D} = \int_{\partial D} U^T V dA$ is an integral over the boundary of the domain. The objective in the analytic approach is to convert the primal problem using integration by parts into an equivalent adjoint problem

$$\begin{array}{rlll} \text{Determine} & J & = (Z, f)_D + (C^* Z, e)_{\partial D} & \\ \text{given that} & L^* Z & = g & in\ D \\ & B^* Z & = h & on\ \partial D \end{array} \tag{8}$$

where L^* is the linear PDE which is adjoint to L. B and B^* are boundary condition operators for the primal and adjoint problems, respectively, and C and C^* are (possibly differential) weight boundary operators used in the functionals; these four operators may have different dimensions on different parts of the boundary.

Comparing the discrete adjoint to the continuous adjoint reveals that although the discrete adjoint solves a linear system of equations, the continuous adjoint is an independent PDE which needs to be discretized and solved by itself. It implies that boundary conditions should be set for the continuous adjoint problem and those boundary conditions are dependent on the functional.

Consider the linearized steady-state Euler equations:

$$LU = \frac{\partial}{\partial x}(A_x U) + \frac{\partial}{\partial y}\left(A_y U\right) = 0 \tag{9}$$

where $A_x = \frac{\partial F_x^c}{\partial U}$ and $A_y = \frac{\partial F_y^c}{\partial U}$ are the convective flux Jacobians and $U = \left(\rho\ \rho u_x\ \rho u_y\ \rho E\right)^T$ is the conserved solution vector. Equation 9 is multiplied by Z, integrated over the domain D and integrated by parts to obtain the continuous adjoint equation:

$$L^* Z = -A_x^T \frac{\partial Z}{\partial x} - A_y^T \frac{\partial Z}{\partial y} = 0. \tag{10}$$

We calculated the continuous adjoint fluxes by the Lax–Friedrichs method. The Lax–Friedrichs method is based on the maximum eigenvalue of the Euler Jacobian. The main advantage of this flux function compared to the Roe's flux function is its simplicity. Use of the Lax–Friedrichs flux function greatly simplifies the dissipation term in the flux, reduces computational cost and improves robustness through additional dissipation. It should be mentioned that the fact that we are using two different methods for the primal problem and the continuous adjoint problem does not affect the performance behavior comparison between

the discrete and continuous adjoint problem. That is because the correction is based on the same discrete primal problem for both discrete and continuous adjoint problems.

The boundary operators are different for supersonic and subsonic inflow and outflow. Subsonic inflow/outflow boundary conditions are described by Hayashi et al. [27]. The common output functionals in aerodynamics are lift and drag and therefore the functionals are defined on the walls.

$$J = \int_{\partial D_{wall}} p\hat{n} \cdot \vec{\psi} ds \tag{11}$$

where $\psi = (\cos\alpha, \sin\alpha)^T$ for drag and $\psi = (-\sin\alpha, \cos\alpha)^T$ for lift, and α is the angle of attack. Using the primal boundary operator for the slip wall boundary condition $\vec{u} \cdot \hat{n} = u_x n_x + u_y n_y = 0$ and multiplying by the adjoint solution,

$$\left(\left(n_x A_x + n_y A_y\right) U, Z\right)_{\partial D} = \left(\begin{bmatrix} 0 & pn_x & pn_y & 0 \end{bmatrix}^T, \begin{bmatrix} z_1 & z_2 & z_3 & z_4 \end{bmatrix}^T\right) = pn_x z_2 + pn_y z_3$$

and comparing to the functional of interest, Eq. 11, we obtain the continuous adjoint boundary condition as

$$n_x z_2 + n_y z_3 = \hat{n} \cdot \vec{\psi}. \tag{12}$$

Consider the linearized steady-state Navier–Stokes equations:

$$\begin{aligned} LU = &\frac{\partial}{\partial x}\left(A_x U - D_{xx}\frac{\partial U}{\partial x} - D_{xy}\frac{\partial U}{\partial y}\right) \\ &+ \frac{\partial}{\partial y}\left(A_y U - D_{yx}\frac{\partial U}{\partial x} - D_{yy}\frac{\partial U}{\partial y}\right) = f \end{aligned} \tag{13}$$

where A_x and A_y are as described above and the other D_{ij} matrices are the derivatives of viscous flux with respect to the solution gradients. Equation 13 is multiplied by Z, integrated over the domain D and integrated by parts to obtain the continuous adjoint equation:

$$\begin{aligned} L^* Z = &- A_x^T \frac{\partial Z}{\partial x} - A_y^T \frac{\partial Z}{\partial y} \\ &- \frac{\partial}{\partial x}\left(D_{xx}^T \frac{\partial Z}{\partial x} + D_{yx}^T \frac{\partial Z}{\partial y}\right) - \frac{\partial}{\partial y}\left(D_{xy}^T \frac{\partial Z}{\partial x} + D_{yy}^T \frac{\partial Z}{\partial y}\right) \end{aligned}$$

The most important target quantities in viscous compressible flows are the total (i.e., the pressure induced plus viscous) drag and lift. So, the output functional is defined as:

$$J = \int_{\partial D_{wall}} \left(p\hat{n} - \tau\hat{n}\right) \cdot \vec{\psi} ds = \int_{\partial D_{wall}} \left(pn_i - \tau_{ij} n_j\right) \psi_i ds \tag{14}$$

where ψ is the same as defined above and τ is the viscous stress term. Using the primal boundary operator for adiabatic no-slip wall boundary condition, $u = v = 0$ and $\hat{n} \cdot \nabla T = 0$, and multiplying by the adjoint solution, we obtain the continuous adjoint boundary condition for the adiabatic no-slip wall:

$$Z_2 = \psi_1, \;\; Z_3 = \psi_2, \;\; \hat{n} \cdot \nabla Z_4 = 0 \tag{15}$$

4 Output Functional Correction

As described by Pierce and Giles [5], the output functional is a function of the truncation error and the adjoint solution. An overview of the method is discussed here.

$$\begin{aligned} J_h &= (g, U_h + (U - U_h))_D + (h, C\,(U_h + (U - U_h)))_{\partial D} \\ &= \left(L^* Z_h, U_h\right)_D + \left(L^* Z_h, U - U_h\right)_D + \left(g - L^* Z_h, U - U_h\right)_D + \left(g - L^* Z_h, U_h\right)_D \\ &\quad \left(B^* Z_h, C U_h\right)_{\partial D} + \left(B^* Z_h, C\,(U - U_h)\right)_{\partial D} + \left(h - B^* Z_h, C\,(U - U_h)\right)_{\partial D} \\ &\quad + \left(h - B^* Z_h, C U_h\right)_{\partial D} \end{aligned} \tag{16}$$

By using integration by parts, we can re-write the above equation as

$$\begin{aligned} J_h &= \left(L^* Z_h, U_h\right)_D + \left(B^* Z_h, C U_h\right)_{\partial D} \\ &\quad + (Z_h, L\,(U - U_h))_D + \left(C^* Z_h, B\,(U - U_h)\right)_{\partial D} \\ &\quad + (Z - Z_h, L\,(U - U_h))_D + \left(C^*\,(Z - Z_h)\,, B\,(U - U_h)\right)_{\partial D} \end{aligned} \tag{17}$$

The two terms in the first line are the influence of bulk and boundary integrals in the discrete functional, the terms in the second line represent computable adjoint error estimates and the two terms in the third line are the higher order remaining errors which are negligible compared to other terms. Accordingly, the estimated error in calculating the functional is:

$$\begin{aligned} \delta J_h &= (Z_h,\, f - L U_h)_D + \left(C^* Z_h, e - B U_h\right)_{\partial D} \\ &= (Z_h, \tau)_D + \left(C^* Z_h, e - B U_h\right)_{\partial D} \end{aligned} \tag{18}$$

where τ is the truncation error. Hence, an estimate of the truncation error is needed to find the correction term for the output functional. The estimate of the truncation error is calculated using higher order flux integral based on second order solution. Details of obtaining the estimate of the truncation error using the higher order flux integral based on the second order solution is described by Sharbatdar [15]. Both the primal and adjoint problems are solved to second-order, in contrast to the finite element method where the higher order, higher than the second-order, adjoint solution is required. Therefore, the correction term is calculated using the second order primal and adjoint solutions and the only higher order operator we use is the higher order flux integration for which the cost is considerably less than obtaining the higher order solution. Moreover, the truncation error and the adjoint solution used to find the correction term are calculated on the same mesh, rather than the refined mesh. Accordingly, we do not need the solution with higher accuracy or a finer mesh.

5 Mesh Adaptation

Adjoint-based adaptation localizes mesh refinement to those parts of the computational domain that most influence the accuracy of the selected output functional of interest [26], optimally reducing error in an output functional by refining the mesh in places where the PDE is poorly solved and that error has a large impact on the output of interest. For situations where the only desired output is a function of the solution, adjoint-based methods are ideal because they do not refine areas of the mesh where the solution has no effect on that function. Analyzing the drag force on an airfoil is an example; the mesh created by adjoint-based refinement provides more accurate drag prediction than other meshes of similar size. The indicator we are using targets areas of large primal truncation error weighted by the adjoint

solution. That is, the correction term in Eq. 18 which is used as the adaptation indicator for adjoint-based mesh adaptation.

The equidistribution of error, or the weighted error, is the principle of all mesh adaptation strategies [28]. The objective of equidistribution is defined as finding a mesh such that the integral of the weight function defined on a domain takes a given constant value over each element or such that the weight function is equidistributed over the chosen mesh. For mesh adaptation, we need to set a length scale for each vertex and the length scale is calculated based on the adaptation indicator. Assuming the correction term in Eq. 18, δJ is chosen as the adaptation indicator, corresponding to the adjoint-based adaptation. We have chosen adjoint-based mesh adaptation using the correction term as by finding the adjoint solution and the truncation error, this adaptation indicator can be obtained directly without any other calculation. Another common option used as an adaptation indicator considers the higher order correction term for the output functional [8]. However, this term requires an estimate of the error in the adjoint solution itself, which in turn requires either a finer mesh solution for the adjoint, or solving an error transport equation for the error in the adjoint solution. Both of these add significant computational cost and complication compared with using only the leading-order correction term, which we will show is still highly effective. Similar to Sect. 4, both the primal and adjoint problems are solved to second-order.

So the correction term δJ is used as the adaptation indicator. The average error is defined as:

$$\overline{\delta J}_h = \frac{\sum_{i=1}^{N} A_i \delta J_i}{\sum_{i=1}^{N} A_i} \tag{19}$$

where A_i is the area of each cell and N is the total number of cells. The target area of each cell is set as:

$$A_i^* = \frac{1}{2} \frac{\overline{\delta J}_h}{\delta J_i} A_i \tag{20}$$

which reduces cell area by a factor of 2, on average. The length scale for regenerating the unstructured mesh using a Delaunay mesh algorithm is defined as:

$$l_i^* = C_l \sqrt{A_i^*} \tag{21}$$

where $C_l = \sqrt{3}$. Using the length scale, the criteria for the element size and quality are met. The length scale is set for each vertex. Using the length scale for each vertex, we need to transform the adaptation indicator assigned to each cell to vertices. For isotropic meshes, the length scales associated with all cells sharing a vertex are averaged and the average value is assigned to each vertex. For generating the initial mesh and the adapted meshes, the GRUMMP meshing library is used [29].

6 Truncation Error Estimation and Smoothness

The exact continuous solution of the problem, if it is known, can be used in the discrete residual which provides us the exact truncation error. If the exact solution is unknown, as for almost all applied problems, the higher order discretization can be used to estimate the truncation error. This method is called the p-TE estimate. This approach has been used for

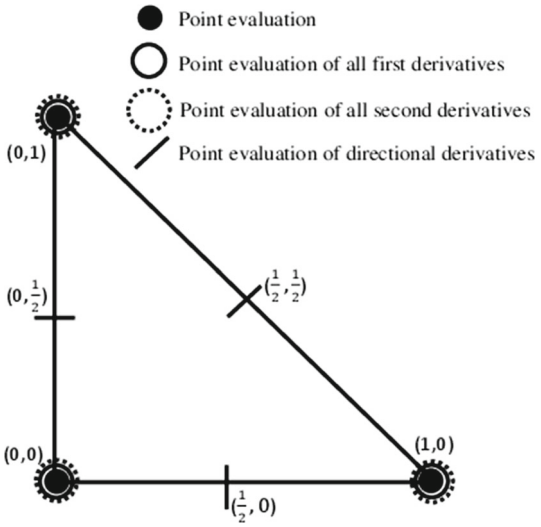


Fig. 1 Argyris reference element

truncation error estimation for structured meshes [6], but behaves in a completely different way on unstructured meshes.

For structured meshes, the truncation error associated with the leading-order term in the discretization error is equal to the higher-order flux integral based on lower order solution. Therefore, p-TE method for estimating the truncation error works properly. However, truncation error for the unstructured mesh finite volume method is asymptotically larger than the discretization error and contains a large amount of noise. The rough modes, based on eigenanalysis using the Jacobian matrix, are the dominant modes in the truncation error [14]. So, we developed a smooth estimate of the truncation error to overcome the discontinuities of the fluxes in the neighboring cells. We use a C^1 interpolation of the solution to evaluate the analytic fluxes and integrate them to compute a truncation error estimate. To compute the C^1 interpolation, we to use the Argyris element [30,31] because of its high rate of spatial convergence [$O(h^6)$ in the L_2 norm for sufficiently smooth problems] and simplicity of implementation compared to other elements. The element is shown in Fig. 1. We reconstruct a discrete solution of order p to higher order, $p+1$, to obtain values at vertices and edges. Details of C^1 interpolation of the solution are described in [15]. By using this C^1 interpolated solution, a unique solution and gradient are obtained at quadrature points of two neighboring cells. We estimate truncation error by integrating the weak form of the PDE around each control volume. That is, we compute

$$\tau_i = \oint \vec{F}_p^{p+1} \cdot \hat{n} ds \tag{22}$$

using Gauss quadrature. The higher-order flux integration can be replaced by $p+2$ as well.

The estimate of the truncation error based on the smoothed solution is multiplied by the adjoint solution to find the correction term added to the output functional to have a more accurate functional which converges to the exact value with the higher-order rate. It should be noted that we compute the correction term of Eq. 18 as the summation over all control volumes of the truncation error calculated by higher-order flux integral based on lower-order solution multiplied by the adjoint solution of the same order as the primal problem

$$\delta J_h = (Z_h, \tau)_D = \sum_{CV} \left(\oint \vec{F}_p^{p+1} \cdot \hat{n} ds \right) \bar{Z}_p$$

7 Results and Discussion

As mentioned, the effectiveness of our smoothing scheme in correcting the output functional has been shown for both convective and diffusive fluxes with manufactured solutions [15]. Furthermore, the functional correction (for both drag and lift functionals) based on the smoothed truncation error was successfully implemented for subsonic inviscid flow on NACA 0012 airfoil. In this paper, inviscid and viscous flows are considered. Because these test cases do not have exact solutions, the solution on a very fine mesh, much finer than the finest mesh in each case, is considered as the exact functional.

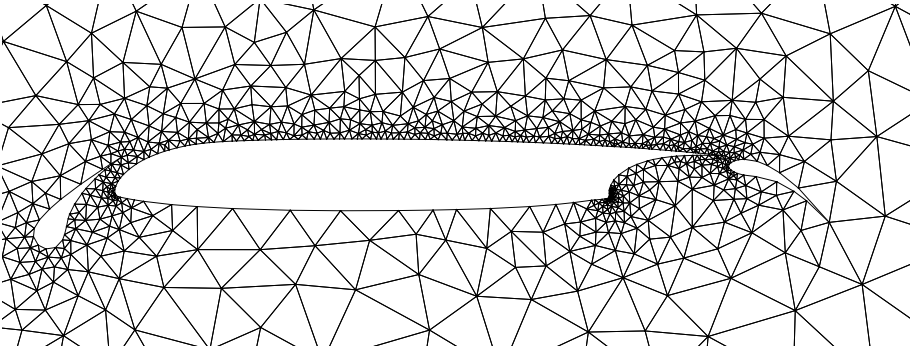
Before showing the numerical results, the nomenclature we are using should be described; J is the original non-corrected functional, CJ is the corrected functional and C's superscript represents whether the correction is based on truncation error calculated by using the non-smooth discrete solution, C^n, or is based on the truncation error calculated based on the C^1 interpolation of the solution, C^s. J's subscript shows the primal solution order of accuracy and the superscript shows the order of reconstruction and flux integration used in estimating the truncation error.

7.1 Inviscid Subsonic Flow on Multi-element Airfoil, $M_\infty = 0.2, \alpha = 3°$

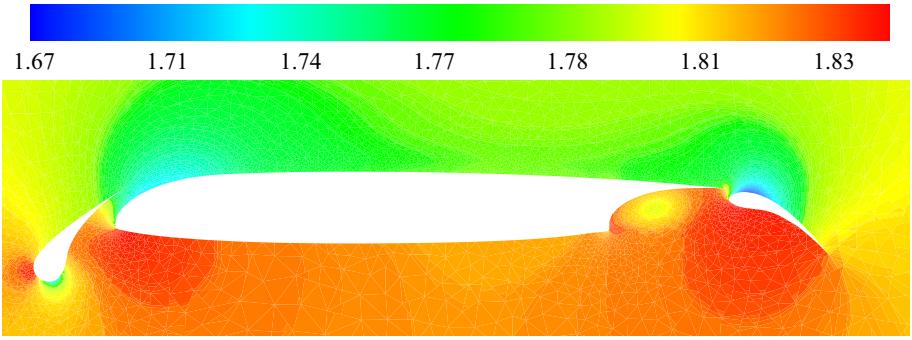
As the effectiveness of our smoothing scheme in correcting the output functional has been investigated for inviscid subsonic flow around a NACA 0012 airfoil [15], we consider a more complicated geometry, a multi-element airfoil, in this section. The boundary of the multi-element airfoil is described in the attachment. The second-order solution on a mesh with characteristic cell size eight times smaller in area than the finest mesh for which correction is applied is considered as the exact solution to the problem. The initial mesh consists of 1877 cells is shown in Fig. 2a. The pressure distribution for the primal problem is also depicted in Fig. 2b.

As mentioned earlier, for correcting the functional either the continuous or discrete adjoint solution can be used. Both continuous and discrete adjoint solutions for the lift functional are depicted in Fig. 3. As these multi-element airfoils are high-lift configuration designs, the lift is considered as the output functional of interest. Although the adjoint solutions are qualitatively similar, the continuous adjoint is smoother, as expected, as a consequence of solving the continuous adjoint PDE instead of using the transpose of the Jacobian arising from the discrete solution. This increased accuracy is reflected in the correction process as well.

The error in the corrected functional is depicted in Fig. 4 using both continuous and discrete adjoint solutions. Comparing the functional based on second-order solution with the corrected functional based on third order flux integration using the non-smoothed discrete solution shows poor performance of the correction in terms of asymptotic convergence rate for both continuous and discrete adjoint solutions. On the other hand, corrected functional by using the truncation error based on the smoothed solution, $C^s J_2^3$, has smaller error compared to the second-order functional. The quartic and cubic convergence is obtained for continuous and discrete adjoint, respectively. In this case, we have super-convergence with the continuous adjoint and the convergence is even better than quartic instead of cubic. It should be noted

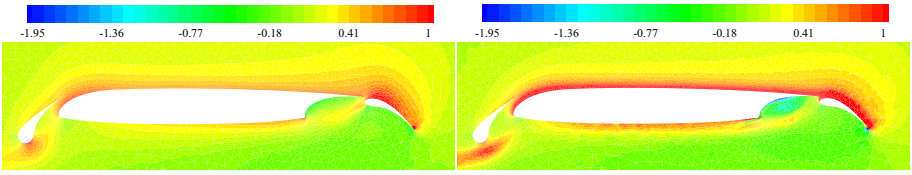


(a)



(b)

Fig. 2 Initial mesh and the primal solution distribution for subsonic inviscid flow on the multi-element airfoil, $Ma_\infty = 0.2, \alpha = 3^\circ$. **a** Close-up of multi-element airfoil mesh consisting of 1877 cells, **b** pressure distribution



(a) **(b)**

Fig. 3 Energy component of the adjoint solution for subsonic inviscid flow on the multi-element airfoil, $Ma_\infty = 0.2, \alpha = 3^\circ$. **a** Continuous adjoint for lift functional, **b** discrete adjoint for lift functional

that the convergence rates shown in these figures are the slopes of the best line which fits the four points using the least-squares method.

Since the same range is used for errors for both continuous and discrete adjoints, the magnitudes of the errors may be compared easily and it is obvious that for all cases, the magnitude of the error in the corrected functional based on the continuous adjoint is less than the corresponding value for the discrete adjoint. This is expected behavior as a result of solving the continuous adjoint PDE. Furthermore, the convergence rate is faster using the continuous adjoint. In conclusion, significant improvement is obtained by correcting the

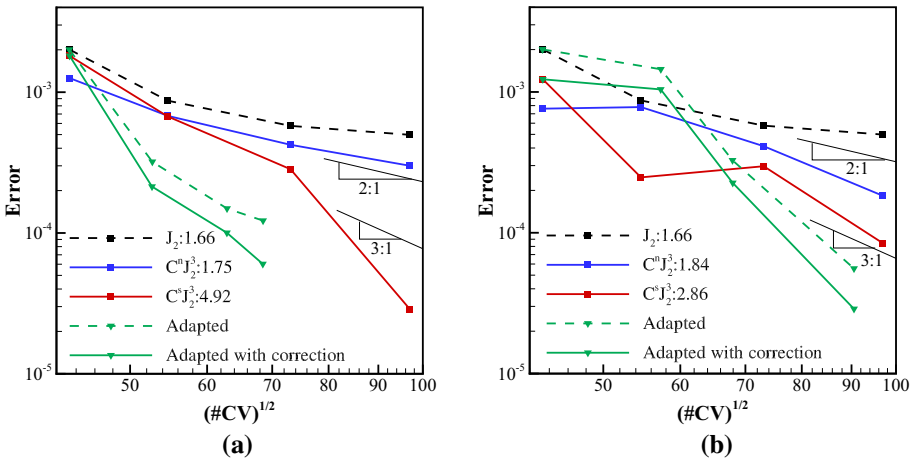


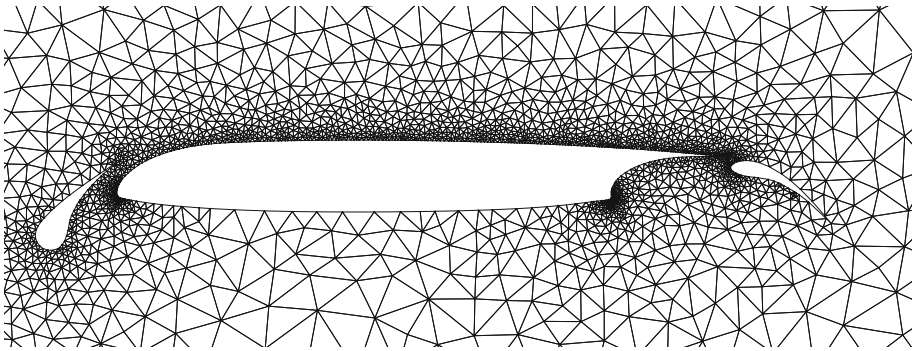
Fig. 4 Convergence history for subsonic inviscid flow on the multi-element airfoil, $Ma_\infty = 0.2$, $\alpha = 3°$. **a** Continuous adjoint for lift functional, **b** discrete adjoint for lift functional

functional by the truncation error based on the smoothed solution, whether the continuous or discrete adjoint is used, although the continuous adjoint is more successful.

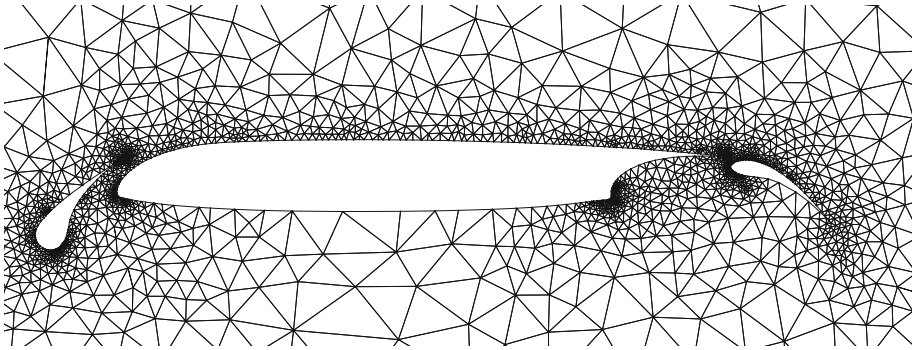
The green line with triangle symbols in these plots shows the results for the adapted meshes. The adaptation indicator is the smoothed truncation error weighted by the adjoint solution. The dashed line is the second-order non-corrected functional calculated on the adapted mesh and the solid line is the corrected functional on the adapted mesh. The correction term uses the third-order flux integral based on the smoothed second-order solution multiplied by the adjoint solution. Two points needs to be considered here; first, adapting the mesh based on the smoothed estimate of the truncation error weighted by the adjoint solution provides a considerably more accurate functional compared to the uniformly refined mesh, and second, correcting the functional calculated on the adapted mesh provides us even a more accurate value which converges to the exact value faster.

A uniformly refined mesh and an adapted mesh with approximately the same number of control volumes are shown in Fig. 5. The adapted mesh is refined based on the smoothed truncation error weighted by the continuous adjoint solution. Although on the uniformly refined mesh, all cells around the solid wall get refined, the adapted mesh focuses mainly on the leading and trailing edges of each element, the regions where the lift is affected significantly.

By estimating the truncation error based on a C^1 interpolated smoothed solution, higher order convergence of the functional is obtained by using the lower order solution. Hence, paying the extra cost of computing the higher-order solution to obtain a more accurate functional which converges to the exact value with higher-order rate is not required and the higher-order convergence is obtained. Smoothing the solution using C^1 interpolation, however, is an extra computational cost compared to the lower-order solution. Figure 6 compares the CPU time for solving the second-order primal problem, the second-order continuous adjoint and the discrete adjoint problems used for correction and the CPU time required to obtain the C^1 interpolation of the solution for finding the smoothed estimate of the truncation error. Both the primal and continuous adjoint problems are solved to second-order using the same reconstruction scheme. The convergence process terminates when the L_2 -norm of the absolute residual vector drops below 4×10^{-9}. Four meshes shown in this figure are uniformly



(a)



(b)

Fig. 5 Uniformly refined and adapted meshes for subsonic inviscid flow on the multi-element airfoil, $Ma_\infty = 0.2$, $\alpha = 3^\circ$. **a** Uniformly refined mesh (number of cells ≈ 5000), **b** adapted mesh (number of cells ≈ 5000) for the lift functional

refined meshes and are the same as the meshes we are using for functional correction in Fig. 4. Time comparison shows that the post processing time required for C^1 interpolation of the solution is small compared to the solution time, and scales linearly with problem size. For discrete adjoint solution, the transpose of the Jacobian matrix is calculated based on converged second-order solution and the single linear system is solved via GMRES method preconditioned by ILU(2). Sensitivity of the functional with respect to the primal solution, or the right-hand side of the Eq. 6, is only a function of pressure in the case of inviscid flow. Therefore, the time required for solving the linear system to obtain the discrete adjoint solution is significantly smaller than the time required to solve the continuous adjoint PDE for the inviscid case where there is not any viscous term which enhances the stability of the problem. For this inviscid case, the residual for the continuous adjoint solver oscillates before getting close to the solution, after which a larger time step can be used to obtain the converged solution. The oscillatory convergence behavior of the primal and the continuous adjoint problems for the second finest mesh is shown in Fig. 7. The number of iterations are of the same order for the two problems.

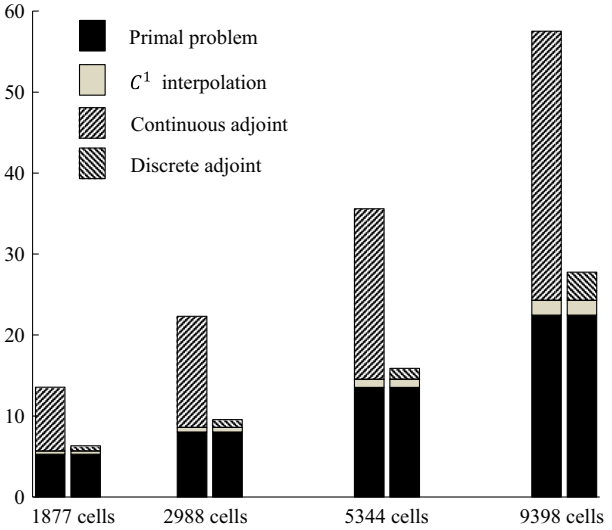


Fig. 6 CPU time comparison (seconds) for primal and adjoint solutions and C^1 interpolation of the solution for subsonic inviscid flow

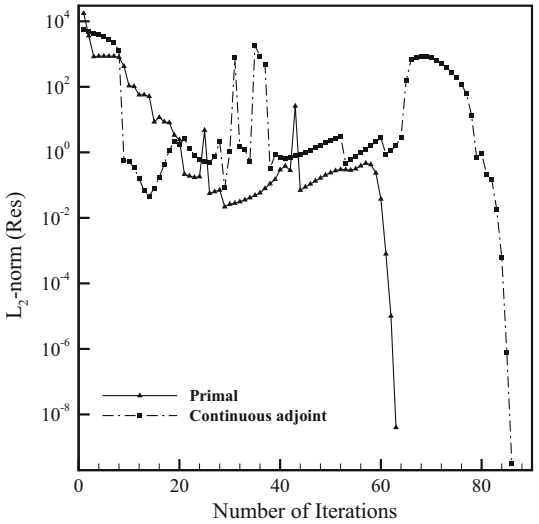
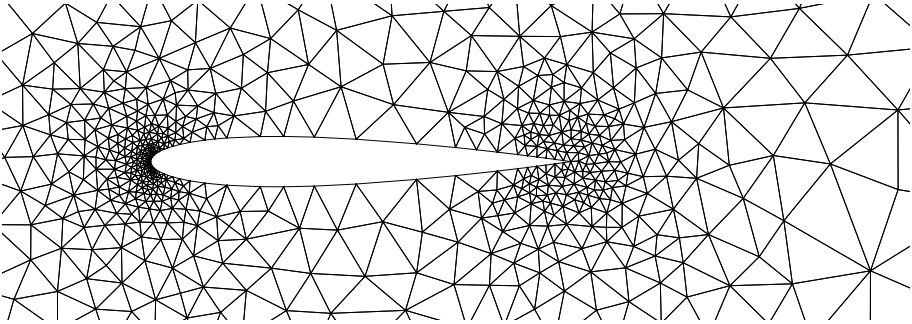


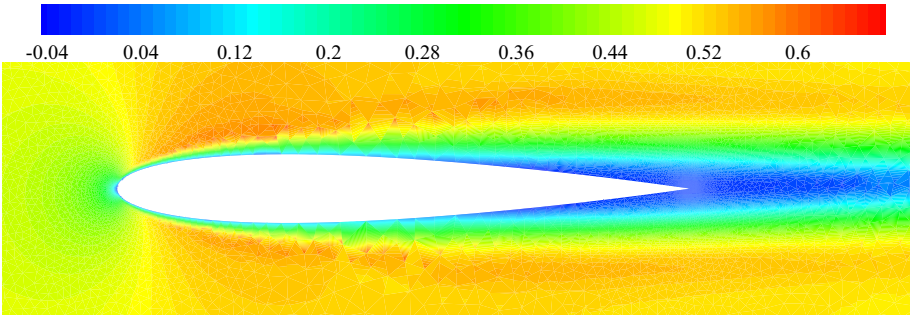
Fig. 7 Convergence behavior of the primal and continuous adjoint problems for subsonic inviscid flow on the multi-element airfoil

7.2 Viscous Subsonic Flow on NACA 0012 Airfoil, $Ma_\infty = 0.5, \alpha = 1°, Re = 5000$

We consider viscous subsonic flow around the NACA 0012 airfoil with a free-stream Mach number of $Ma_\infty = 0.5$, a Reynolds number of $Re = 5000$, and an angle of attack $\alpha = 1°$. The second-order solution on a mesh with characteristic cell size thirteen times smaller in area than the finest mesh for which correction is applied is considered as the exact solution to the problem. The farfield boundary is a circle of radius 100 chords centered at the leading

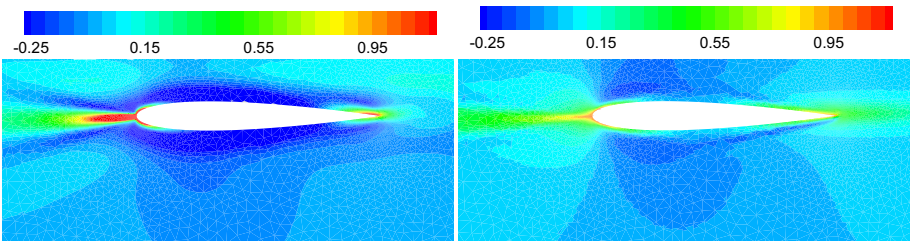


(a)



(b)

Fig. 8 Initial mesh and the primal solution distribution for subsonic inviscid flow on NACA 0012 airfoil, $Ma_\infty = 0.5$, $\alpha = 1°$, $Re = 5000$. **a** Close-up of NACA 0012 airfoil mesh consisting of 2617 cells, **b** x-momentum distribution



(a) **(b)**

Fig. 9 x-Momentum component of the adjoint solution for subsonic viscous flow on NACA 0012 for the drag functional, $Ma_\infty = 0.5$, $\alpha = 1°$, $Re = 5000$. **a** Continuous adjoint for drag functional, **b** discrete adjoint for drag functional

edge of the the airfoil and consequently, the errors arising from boundary placement can be neglected. The initial mesh consists of 2617 cells is shown in Fig. 8a. The x-momentum distribution for the primal problem is also depicted in Fig. 8b. Continuous and discrete adjoint solutions for the same component are depicted in Fig. 9. As viscous flow is studied, the total drag is considered as the output functional. Both adjoint solutions have the same distribution, qualitatively; however the continuous adjoint solution has a smoother distribution.

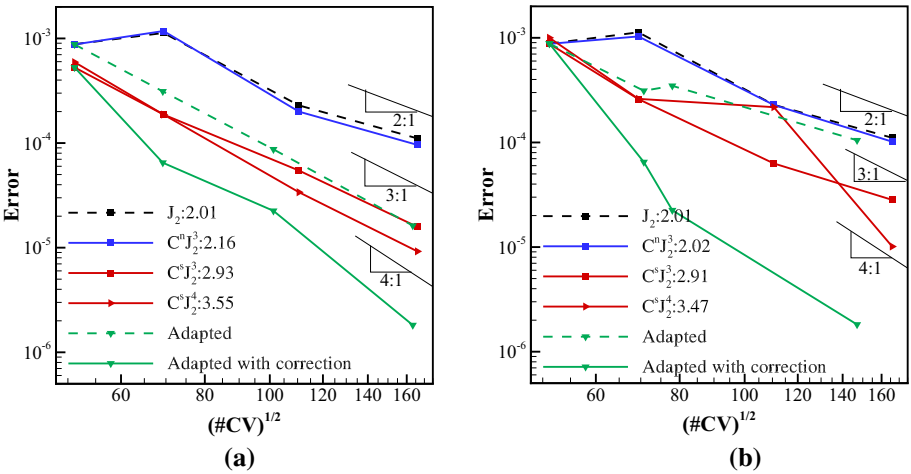


Fig. 10 Convergence history for subsonic viscous flow on NACA 0012, $Ma_\infty = 0.5$, $\alpha = 1°$, $Re = 5000$. **a** Continuous adjoint for drag functional, **b** discrete adjoint for drag functional

Functional correction based on these adjoint solutions is applied and the results are shown in Fig. 10. Similar to the subsonic inviscid case, comparing the black line representing the original second-order solution with the blue line representing the corrected functional using the non-smoothed truncation error shows that correction based on the non-smoothed truncation error is not helpful either in terms of the error magnitude or the convergence rate. However, by correcting the functional using the smoothed truncation error cubic convergence is obtained for both continuous and discrete adjoint solutions. If fourth-order flux integral (red line with triangle symbols) is used instead of third order flux integral (red line with square symbols), the corrected functional converges quartically, approximately. Therefore, quartic convergence is obtained by just solving the primal problem at second-order. It should be noted that although functional correction using finite-element methods requires a higher-order adjoint solution to obtain higher-order convergence, we are using the adjoint problem of the same order as the primal problem and we obtain higher-order convergence using the finite-volume method. Similar to the previous inviscid test case, the convergence rates shown in these figures are the slopes of the best line which fits the four points using the least-squares method.

The green line with triangle symbols shows the error in calculating the functional on the adapted mesh and the adaptation indicator is again the smoothed truncation error weighted by the adjoint solution. As the effect of correction on the adapted mesh was investigated investigated in Sect. 7.1, we only show the corrected functional on the adapted mesh in this part. It is obvious that the adapted mesh outperforms the uniformly refined mesh.

The uniformly refined and the adapted meshes with approximately the same number of cells are shown in Fig. 11. Although the uniformly refined mesh focuses only on the leading and trailing edge, the adapted mesh has enough resolution along the boundary layer and the wake region and consequently, a more accurate functional is obtained. The farfield view shows that the adapted mesh only focuses on regions near the solid wall compared to the uniformly refined mesh with approximately the same number of cells in which the farfield is refined as well.

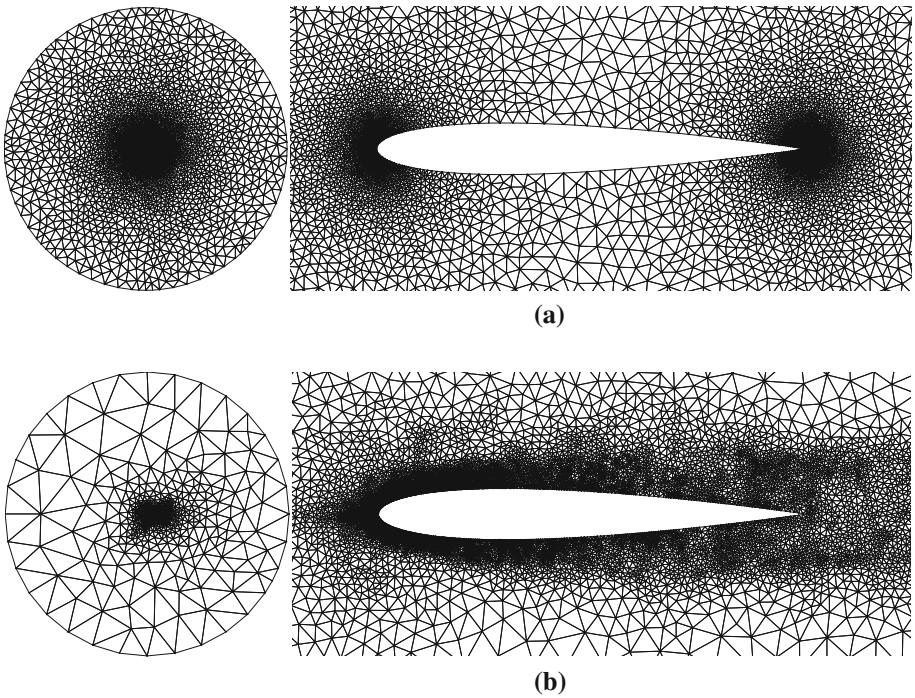


Fig. 11 Farfield (left) and close-up (right) view of the uniformly refined and adapted meshes for subsonic viscous flow on NACA 0012, $Ma_\infty = 0.5$, $\alpha = 1°$, $Re = 5000$. **a** Uniformly refined mesh (number of cells ≈ 26000), **b** adapted mesh (number of cells ≈ 26000) for the drag functional

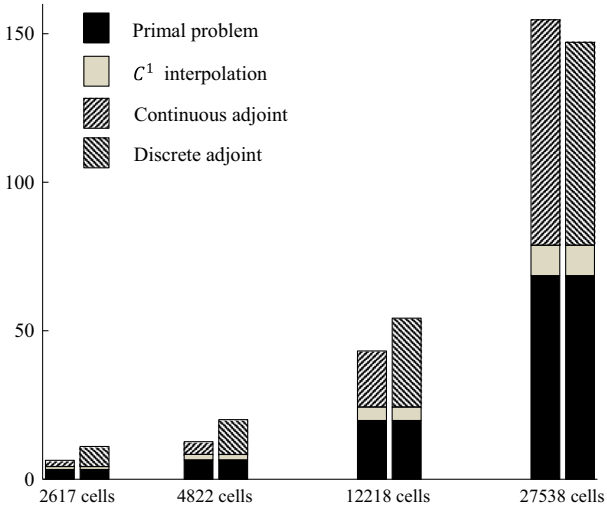


Fig. 12 CPU time comparison (seconds) for primal and adjoint solutions and C^1 interpolation of the solution for subsonic viscous flow

The CPU time required to solve the second-order primal problem, the second-order continuous and the discrete adjoint problems used for correction and the time for obtaining the C^1 interpolation of the solution for this subsonic viscous test case is shown in Fig. 12. Similar

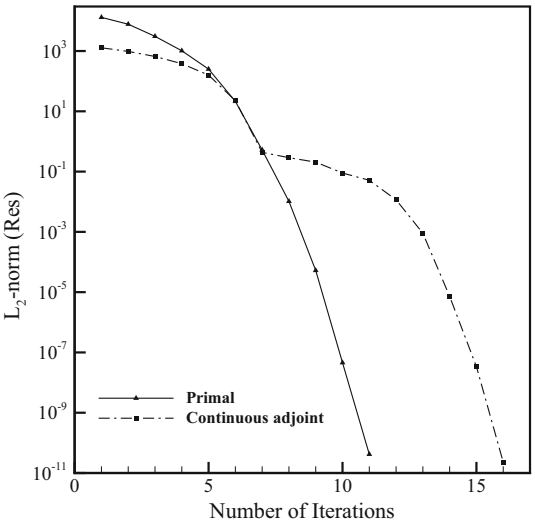


Fig. 13 Convergence behavior of the primal and continuous adjoint problems for subsonic viscous flow on NACA 0012 airfoil

to the inviscid case, the time required for obtaining the C^1 interpolation of the solution is much smaller the time for solving the primal problem. However, the time for solving the continuous and discrete adjoint solutions are comparable. As a consequence of the viscous terms, convergence of the continuous adjoint PDE is faster compared to the inviscid case. Furthermore, the sensitivity of the functional with respect to the solution is more complicated in the case of viscous flow and consequently finding the right-hand side of Eq. 6 to obtain the discrete adjoint solution requires more time. The convergence behavior of the primal and the continuous adjoint problems for the second finest mesh is shown in Fig. 13. The number of iterations are of the same order for the two problems and they are considerably smaller compared to the inviscid test case, Fig. 7.

Another point that needs to be considered here is the adjoint consistency for the discrete adjoint solution. We show adjoint consistency for this test case which involves both convective and diffusive terms. The L_2-norm of the difference between the discrete and continuous adjoint solutions is shown in Fig. 14. The difference between the discrete and continuous adjoint solution decreases with mesh refinement, implying that the discretization scheme is asymptotically adjoint consistent.

7.3 Inviscid Transonic Flow on NACA 0012 Airfoil, $M_\infty = 0.8, \alpha = 1.25°$

Inviscid transonic flow around NACA 0012 airfoil with a free-stream Mach number of $Ma_\infty = 0.8$ and an angle of attack $\alpha = 1.25°$ is considered. The second-order solution on a mesh with characteristic cell size eight times smaller in area than the finest mesh for which correction is applied is considered as the exact solution to the problem. The farfield boundary is a circle of radius 100 chords centered at the leading edge of the the airfoil and consequently, the errors arising from boundary placement can be neglected. The initial mesh consists of 1998 cells is shown in Fig. 15. The pressure distribution for the primal problem is also depicted in Fig. 15.

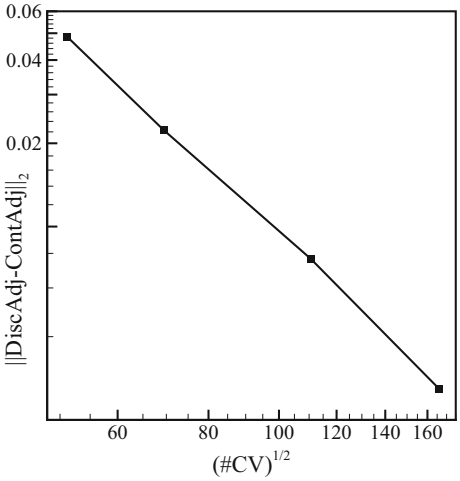


Fig. 14 L_2-norms of the difference between the discrete and continuous adjoint solutions for subsonic viscous flow on NACA 0012 airfoil

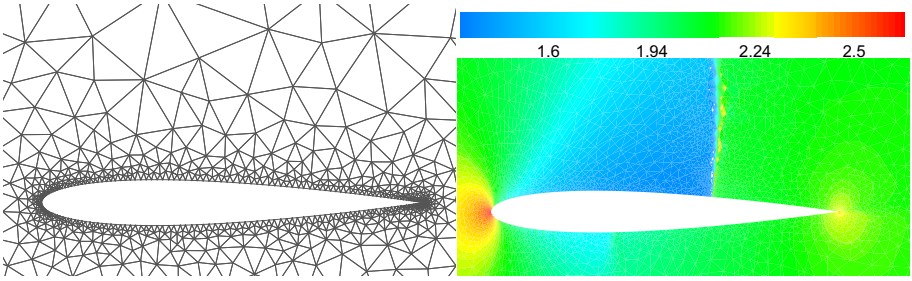


Fig. 15 Initial mesh and the primal solution distribution for transonic inviscid flow on NACA 0012 airfoil, $Ma_\infty = 0.8$, $\alpha = 1.25°$. **a** Close-up of NACA 0012 airfoil mesh consisting of 1998 cells, **b** pressure distribution

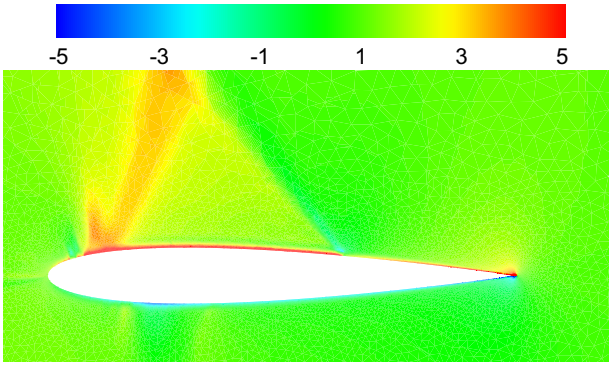


Fig. 16 x-momentum adjoint solution for the lift functional, transonic inviscid flow on NACA 0012 airfoil, $Ma_\infty = 0.8$, $\alpha = 1.25°$

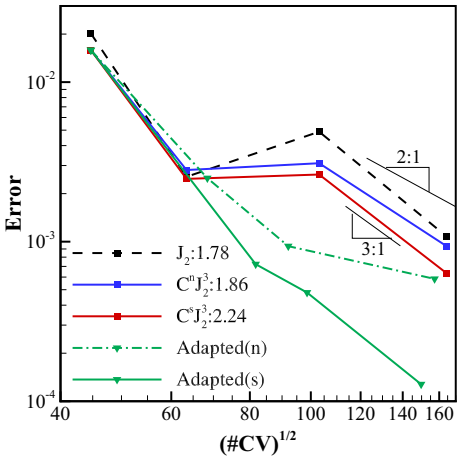


Fig. 17 Convergence history for transonic inviscid flow on NACA 0012 for the lift functional, $Ma_\infty = 0.8$, $\alpha = 1.25°$

The discrete adjoint solution distribution for the lift functional is shown in Fig. 16. For this case, due to some robustness issues for the continuous adjoint solution, we just show the results for the discrete adjoint. Functional correction based on the discrete adjoint solution for the lift functional is depicted in Fig. 17. Functional correction using the smoothed truncation error provides a more accurate functional which converges to the exact value faster compared to the non-smoothed truncation error. Although the convergence rate calculated based on the four points least-squares method is not cubic, it is faster than the non-smoothed truncation error. This is expected behavior given the discontinuity at the shock region.

The green lines show the adapted mesh using the non-smoothed truncation error weighted by the discrete adjoint solution (dash-dotted line) and the smoothed truncation error weighted by the discrete adjoint solution (solid line) as the adaptation indicator. Comparing these two lines shows that adaptation based on the smoothed truncation error outperforms adaption based on the non-smoothed truncation error. Figure 18 compares the uniformly refined mesh with the adapted meshes based on these two indicators. The adapted mesh using the smoothed truncation error has more resolution near the strong shock on the upper surface. Moreover, the mesh resolution near the weak shock on the lower surface is better compared to the adapted mesh using the non-smoothed truncation error.

8 Concluding Remarks

The focus of this paper is on test cases which are interesting from numerical aerodynamics point of view including the subsonic inviscid flow on a multi-element airfoil, subsonic viscous and transonic inviscid flow on NACA 0012 airfoil. It is known that the truncation error on unstructured meshes has larger magnitude compared to structured meshes and the rough modes in the truncation error are dominant. As a result of discontinuous jumps of the coefficients of the leading terms in the Taylor series expansion of the error from one control volume to another, different values are obtained for the solution and gradients at two neighboring cells where the flux integral is calculated. In this paper, we have used the C^1 interpolation scheme developed by Sharbatdar et al. [15] to obtain a continuous solution and gradients at

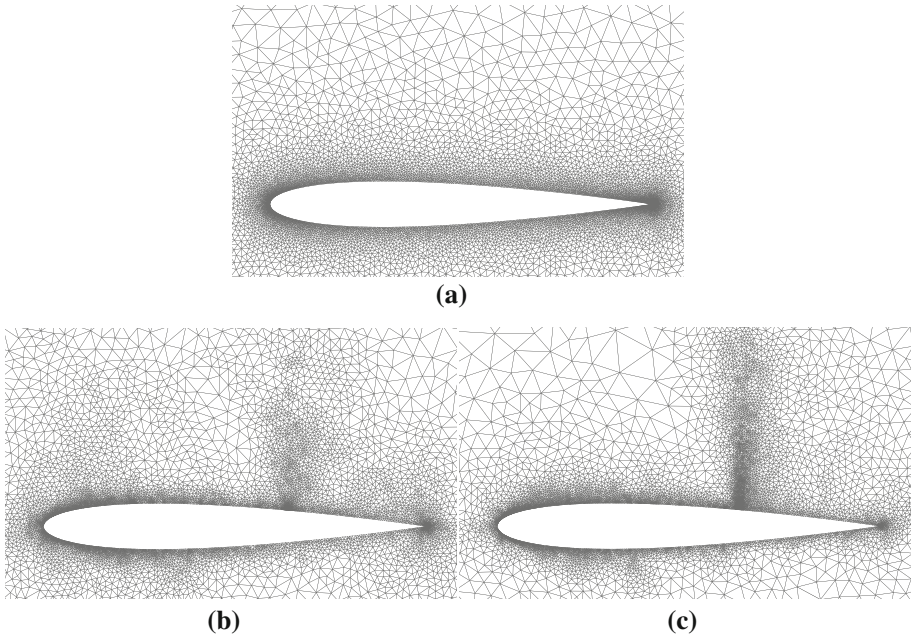


Fig. 18 Uniformly refined and adapted meshes for transonic inviscid flow on NACA 0012 airfoil for the lift functional (number of cells ≈26,000), $Ma_\infty = 0.8, \alpha = 1.25°$. **a** Uniformly refined mesh, **b** adapted mesh by the non-smoothed TE weighted by the discrete adjoint, **c** adapted mesh by the smoothed TE weighted by the discrete adjoint

the quadrature points and to have a single value for the flux vector from two adjacent cells. Since we are using the p-truncation error approach for estimating the truncation error, which computes the higher-order flux integration based on the lower-order solution, the smooth solution and consequently the smooth flux vector is able to provide us a smoother estimate of the truncation error. The truncation error estimate when multiplied by the adjoint solution is used to compute a correction term for an output functional of interest. We combine the smooth truncation error estimate with both continuous and discrete adjoints and compare their performance. The smoothed truncation error weighted by the adjoint solution can also be used to drive mesh adaptation.

Our results show that using a truncation error estimate based on the non-smoothed solution does not improve the value of the functional, either in terms of the convergence rate or in terms of the error magnitude, for all problems we have investigated. Although using the truncation error based on the non-smoothed solution is not helpful, using the C^1 interpolated solution improves the functional accuracy significantly. The second-order solution is used to find a continuous flux integral based on the third or fourth-order reconstruction giving third or fourth-order convergence for the functional while only solving the primal and adjoint problems at second-order. The p-sharp convergence obtained using the smoothed estimate of the truncation error and the adjoint solution is a new property of the finite volume method.

We have shown that the extra cost of obtaining the smooth estimate of the truncation error is negligible compared to the solution time. More accurate corrected functionals are obtained when correction is based on the continuous adjoint since a more accurate PDE is solved compared to solving a linear system which gives the discrete adjoint. As a result,

more consistent and more accurate results are obtained using the continuous adjoint for both inviscid and viscous flows. For the inviscid case, the time required for obtaining the discrete adjoint solution is smaller compared to the continuous adjoint solution; however, these times are approximately similar for the viscous test case.

Significant improvement is obtained when the smoothed truncation error weighted by the adjoint solution is used to drive mesh adaptation. A more accurate functional which converges to the exact value faster is obtained when it is calculated on the adapted mesh, compared to the uniformly refined mesh. Correcting the functional calculated on the adapted mesh improves the value even more. If the smoothed estimate of the truncation error weighted by the adjoint solution is used as the adaptation indicator, the results are more accurate compared to the non-smoothed truncation error and different features of the flow are obtained with higher resolution.

The significant improvement in functional correction or the solution on the adapted mesh is obtained just by smoothing the primal solution and finding the truncation error based on this smoothed solution and then multiplying the truncation error by the adjoint solution of the same order as the primal solution. It is not necessary to smooth the adjoint solution to obtain significant improvement in convergence rates.

The smoothing scheme we developed can be extended to any other two-dimensional problem without any difficulties as it worked properly for both viscous and inviscid flows which involve diffusive and convective terms. Residual-based mesh adaptation using the smoothed estimate of the truncation error may also be more helpful compared to the usual residual-based mesh adaptation. Extending the smoothing scheme to 3D problems involves finding the proper polynomials for the 3D reference element, similar to the Argyris element for 2D, and then solving the system of equations arising from the basis functions once to obtain the constants.

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